# Namit Anand

*Curriculum Vitæ*

*University of Southern California*
*Los Angeles, CA, USA*
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## Research interests

Quantum resource theories, Open quantum systems, Quantifying non-Markovianity, Quantum error-correction, Characterizing quantum resources in quantum algorithms, Many-body theory

## Education

2017 – 2022 (expected) **Doctor of Philosophy, Physics**, *University of Southern California*, Los Angeles, CA, USA. *Advisor*: Paolo Zanardi.

2016 – 2017 **Visiting Student Researcher, Physics**, *Harish-Chandra Research Institute*, Prayagraj, India.

2011 – 2016 **Integrated Master of Science (BS+MS), Physics**, *National Institute of Science Education and Research*, Bhubaneswar, India.

## Publications & Preprints

1. **Quantum coherence in the localization transition**, *Georgios Styliaris, Namit Anand, Lorenzo Campos Venuti, and Paolo Zanardi*, arXiv:1906.09242 [quant-ph] (2019).
2. **Quantifying non-Markovianity: a quantum resource-theoretic approach**, *Namit Anand and Todd A. Brun*, arXiv:1903.03880 [quant-ph] (2019).
3. **Demonstration of fidelity improvement using dynamical decoupling with superconducting qubits**, *Bibek Pokharel, Namit Anand, Benjamin Fortman, and Daniel A. Lidar*, Phys. Rev. Lett. **121**, 220502 (2018).
   – Popular summary of our work in Phys.org, ScienceDaily, and the Scientific American.
   – Featured on the Rigetti website.
4. **Asymmetry and coherence weight of quantum states**, *Kaifeng Bu, Namit Anand, and Uttam Singh*, Phys. Rev. A **97**, 032342 (2018).
5. **Coherence and Entanglement Monogamy in the Discrete Analogue of Analog Grover Search**, *Namit Anand and Arun K. Pati*, arXiv:1611.04542 [quant-ph] (2016).
6. **Comment on "Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics"**, *Namit Anand*, Eur. J. Phys. **37** 048003 (2016).
7. **Do quantum strategies always win?**, *Namit Anand and Colin Benjamin*, Quantum Information Processing **14** (11), 4027-4038 (2015).

- My [Google Scholar] and [arXiv] pages.

## Relevant Graduate Coursework

- Introduction to Quantum Information Processing (*Course Instructor*: Prof. Todd A. Brun)
- Quantum Error Correction (*Course Instructor*: Prof. Daniel A. Lidar)
- Theory of Open Quantum Systems (*Course Instructor*: Prof. Daniel A. Lidar)
- Quantum Information Theory (*Course Instructor*: Prof. Todd A. Brun)
- Quantum Information Science and Many-body Physics (*Course Instructor*: Prof. Paolo Zanardi)

## Teaching Experience

(Fall '17 & Spring '18) Graduate Teaching Assistant for Physics 135aL: Physics for the Life Sciences. Student Evaluation Rating: 5.0/5.0

## Awards and Fellowships

Summer 2018 Recipient of the Quantum computing fellowship at the Los Alamos National Laboratory.

Award: $12,200 (USD)

2017 – 2022 University of Southern California, Department of Physics & Astronomy.

Award: $27,500 (USD) annually; full financial support through teaching assistantship and one year of fellowship for completion of the Ph.D. program.

Summer 2015 One of the five summer fellows (internationally) for the fully funded I.I.A.S.S. Summer internship program, Italy.

Award: €3,000.

2011 – 2016 Recipient of Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship by the Department of Science and Technology, Government of India from 2011 - 2016.

Award: ₹80,000/year for 5 years.

## Undergraduate Research Experience

Aug '16 – May '17 **Identifying Topological Order with Entanglement – the role of quantum information in condensed matter physics**, *Prof. Sumathi Rao*, Harish-Chandra Research Institute.

Sep '16 – May '17 **Concentration of Measure phenomena in quantum information theory**, *Namit Anand and Uttam Singh*, Harish-Chandra Research Institute.

May '16 - '17 **Density Matrix Renormalization Group Algorithms**, *Dr. Anamitra Mukherjee*, NISER.

Aug '15 – '16 **Quantum Games, Walks, and Algorithms**, *Dr. Colin Benjamin*, NISER.

May – July 2015 **Characterizing phases in one-dimensional systems using entanglement**, *Prof. Roberta Citro*, University of Salerno.

Jan – May 2015 **Simulation of two-level quantum system using classical coupled oscillators**, *Dr. Ashok Mohapatra*, NISER.

## Computational Skills

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| Programming Languages | C, C++, Python |
| Specialized Software | Mathematica and MATLAB |
| Markup Languages | LaTeX, HTML, HTML5, PHP |
| Software adept | Adobe Suite (Photoshop, Audition) and Game Development (UDK) |
| Numerical packages | *Lapack* for C++ & Python, MATLAB on HPC, *Quantum Mathematica* for quantum computing simulations in Mathematica, and CVX for convex optimization in MATLAB |
| Quantum simulations | Simulating quantum error-correction on the IBM Quantum Experience |

## Professional References

**Paolo Zanardi**

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University of Southern California
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**Todd A. Brun**

Professor of Electrical Engineering-Systems, Computer Science, and Physics and Astronomy
University of Southern California
Email: tbrun@usc.edu
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**Daniel A. Lidar**

(*Viterbi Professor of Engineering*)
Professor of Electrical Engineering-Systems, Chemistry, and Physics and Astronomy
University of Southern California
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**Arun K. Pati**

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